

Credit Risk Lab

Aditya Gujral | Interactive machine learning case study | October 2026

THE BUSINESS QUESTION

How can an analyst prioritize accounts for review while understanding the tradeoff between capturing future delinquency and flagging accounts that do not experience it? This portfolio application demonstrates that workflow using synthetic data.

THE APPROACH

Generate 6,000 synthetic snapshots and predict simulated transition to 90+ days past due within 90 days. Use seven inputs: starting delinquency, credit score, income, balance, utilization, missed payments and portfolio. Compare logistic regression, random forest and a prior-probability baseline.

Train on Jan–Mar 2025 (1,430 accounts), validate on July (508), and test on Nov–Dec (995). Exclude 3,067 snapshots to allow outcomes to mature before the next split. Fit preprocessing on training data only; keep customer IDs disjoint. Select the model with validation ROC-AUC and its review threshold with validation F2.

VERIFIED RESULTS

Selected model	Logistic regression
Test ROC-AUC	0.721; random forest 0.712; baseline 0.500
Validation-selected threshold	26%
Test precision / recall at 26%	56.3% / 93.0%
At that threshold	480 true positives, 372 false positives, 36 missed events
Browser inference parity	1,495 accounts; maximum error < 1e-12

THE DELIVERABLE

An interactive app with live scoring from both trained models, comparison metrics and ROC curves, a threshold explorer, logistic feature contributions, and a sortable 500-account review queue with CSV export. The 500 scoring accounts are unlabeled; future outcomes are not claimed.

WHAT THE RESULT MEANS

The 26% operating point captures most simulated events but flags many non-events, making the workload tradeoff visible. This is a demonstration of model evaluation and an analyst workflow, not evidence of a real-world recovery uplift. Probabilities are not calibrated or validated against real borrowers. Native browser visual QA remains pending.

Try the public app: <https://aditya-credit-risk-lab.aditya-gujral91.chatgpt.site>

Source, methodology and validation: https://github.com/AadityaGujral/AadityaGujral/tree/main/portfolio/python_credit_risk

Metric reference: https://scikit-learn.org/stable/modules/model_evaluation.html